Variable-time prime-field backends (`fp_backend`)

Study `research/fp_backend_20261015` · 2026-10-15 · module `src/cryptanalysis/fp_backend.rs` · harness `examples/fp_backend_bench.rs`, analysis `examples/fp_backend_report.rs` · protocol `PROTOCOL.md` (committed before the measured run).

What was built

One trait, `PrimeField`, with element arithmetic, inverse, batch inverse, exponentiation, Legendre symbol, square root and a **lazy accumulator** (`F::Wide`: unreduced sums of products, one reduction at the end) with row kernels (`row_load`, `row_sub`, `row_axpy`, `row_scatter_axpy`, `row_reduce`, `row_reduce_scaled`, `dot`). Backends: `Fp32` (canonical `u32`, Barrett; the lazy kernel of `f4_fp_tower` below 2^{31}, Shoup products with a carry fold above), `Fp64` (Montgomery, `p < 2^63`), `FpWide<N>` (Montgomery CIOS, any `p < 2^(64N)`), `FpWide<N, true>` (carry-free CIOS for primes with a spare top bit, alias `Fp224`), `FpP192` (Solinas reduction for NIST P-192). Inverse: Pornin's binary GCD (wide) and Kaliski's almost inverse (one word); Legendre: binary Jacobi; square root: `a^((p+1)/4)` or Tonelli–Shanks with 8/12-bit windowed 2-Sylow logarithms when $v_2(p-1) \geq 16$. Lazy-accumulator capacities are derived in the module documentation and asserted in debug builds: unbounded for `Fp32`/`Fp64` (self-renormalising), `2^64 − 2` products for `FpWide`/`FpP192`.

Correctness

`cargo test --release --lib fp_backend`: 8 tests, every operation of every backend against `BigUint` on edge and random values at 3–17, 31/32-bit, 61–63-bit (two with $v_2(p-1) = 40$), P-192, a 190-bit prime, P-224, P-256, a 200-bit prime, $2^{127}-1$, $2^{128}-159$, one-limb primes to $2^{64}-59$ and a 512-bit prime; lazy rows, scatter rows and dot products against eager reduction including all-(p−1) inputs; the binary GCD never fell back. Same suite passes with debug assertions and overflow checks on. The harness re-checks every timed kernel against `BigUint` or the f4_fp_tower copy before timing.

Run

```
commit: 6979448f00a30c5e29ca7d4d6bf316c69687c2af
command: taskset -c 3 nice -n 0 prlimit --as=4294967296 -- fp_backend_bench --reps 9 --target-ms 40 --out research/fp_backend_20261015/runs/final
start: 2026-10-15T18:13Z
uptime at start:  18:13:04 up  3:16,  0 user,  load average: 1.03, 1.70, 1.33
cpu: Intel(R) Xeon(R) Processor @ 2.10GHz, 4 vCPU
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: shared with another lane, no isolation (L0)
exit: 0
end: 2026-10-15T18:14Z
uptime at end:  18:14:31 up  3:18,  0 user,  load average: 1.04, 1.53, 1.30
```

Wall-clock ns per operation, 9 interleaved repetitions of every arm (≈ 40 ms each), one core (`taskset -c 3`) of a **shared, noisy** 4-vCPU host with another lane running: no isolation (L0). Ratios are paired by repetition (baseline/candidate, > 1 = candidate faster); verdicts on the median paired ratio against the frozen thresholds.

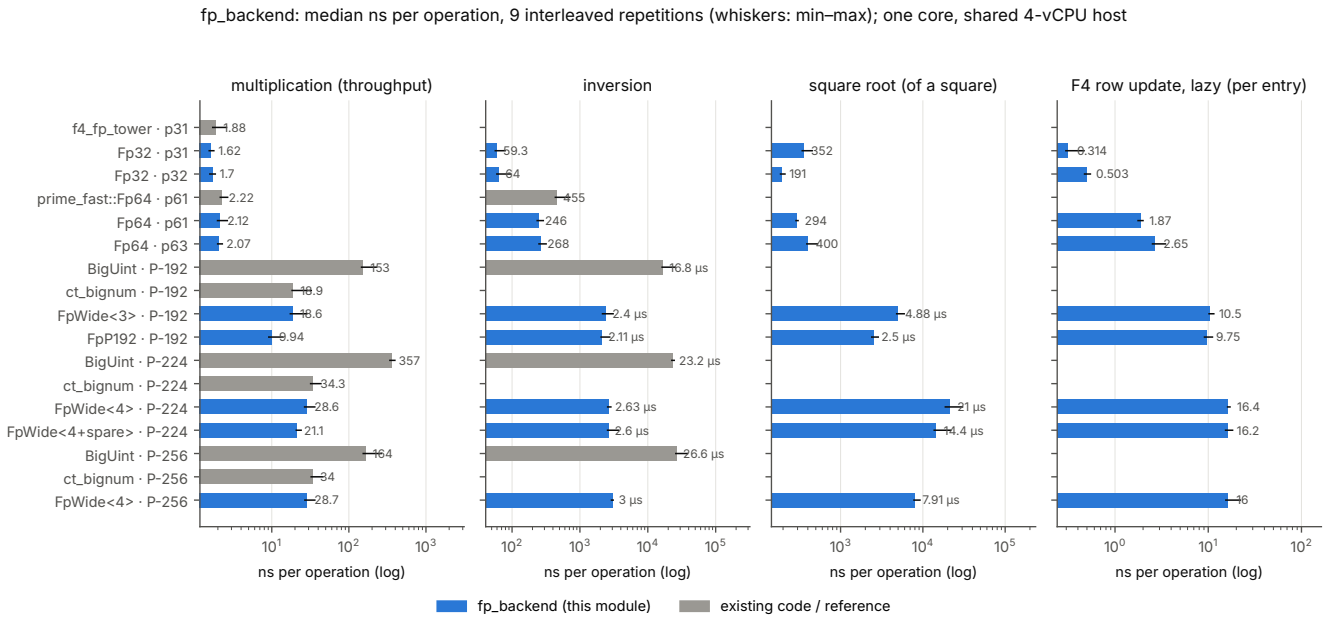


Figure 1: Median ns per operation (whiskers: min–max over 9 repetitions), run `runs/final`. Blue: this module; gray: existing code (`f4_fp_tower` copy, `prime_fast::Fp64`, `ct_bignum`, `BigUint`).

Pre-registered claims

id	claim	base ns	cand ns	ratio	min–max	thr	verdict
C1a	Fp32 dense row kernel, p31, 1e4: no regression vs f4_fp_tower	0.4	0.31	1.308	1.24–1.34	0.95	**PASS**
C1b	Fp32 dense row kernel, p31, 1e5: no regression vs f4_fp_tower	0.44	0.34	1.287	1.18–1.48	0.95	**PASS**
C2a	Fp32 sparse scatter, p31: no regression vs f4_fp_tower	2.03	1.92	1.068	1.03–1.25	0.95	**PASS**
C2b	Fp32 sparse scatter, p32: no regression vs f4_fp_tower	2	1.86	1.067	0.83–1.11	0.95	**PASS**
C3a	Fp32 dense row kernel, p32, 1e4: >= 2x f4_fp_tower (eager above 2^31)	1.26	0.5	2.65	2.4–2.7	2	**PASS**
C3b	Fp32 dense row kernel, p32, 1e5: >= 2x f4_fp_tower	1.33	0.51	2.657	2.26–3.06	2	**PASS**
C4	Fp32 mul throughput, p31: no regression vs f4_fp_tower Fp::mul	1.88	1.62	1.165	1.02–1.5	0.95	**PASS**
C5a	Fp64 mul throughput, p61: no regression vs prime_fast::Fp64	2.22	2.12	1.086	0.86–1.35	0.95	**PASS**
C5b	Fp64 mul latency, p61: no regression vs prime_fast::Fp64	5.35	4.99	1.079	1.07–1.11	0.95	**PASS**
C5c	Fp64 inverse, p61: >= 1.4x prime_fast::Fp64	455.34	245.67	1.871	1.51–2.94	1.4	**PASS**
C6a	FpP192 mul throughput >= 1.5x ct_bignum<3>, P-192	18.86	9.94	1.951	1.51–3.27	1.5	**PASS**
C6b	FpP192 mul latency >= 2x ct_bignum<3>, P-192	30.29	11.15	2.71	2.55–2.94	2	**PASS**
C6c	FpWide<3> mul throughput >= 1.0x ct_bignum<3>, P-192	18.86	18.63	1.035	0.66–1.67	1	**PASS**
C6d	FpWide<3> mul latency >= 1.25x ct_bignum<3>, P-192	30.29	21.93	1.376	1.28–1.5	1.25	**PASS**
C6e	FpWide<4+spare> mul throughput >= 1.3x ct_bignum<4>, P-224	34.26	21.06	1.562	1.5–2.07	1.3	**PASS**
C6f	FpWide<4+spare> mul latency >= 1.8x ct_bignum<4>, P-224	45.17	21.39	2.099	1.36–2.17	1.8	**PASS**
C6g	FpWide<4> mul throughput >= 1.1x ct_bignum<4>, P-256	33.98	28.68	1.256	0.92–1.45	1.1	**PASS**
C6h	FpWide<4> mul latency >= 1.3x ct_bignum<4>, P-256	44.27	29.96	1.477	1.17–1.72	1.3	**PASS**
C7a	FpWide<3> mul >= 5x BigUint (a*b) % p, P-192	152.81	18.63	8.203	7.92–9.93	5	**PASS**
C7b	FpWide<4+spare> mul >= 5x BigUint, P-224	356.77	21.06	16.911	16.22–18.25	5	**PASS**
C7c	FpWide<4> mul >= 4x BigUint, P-256	164.41	28.68	5.42	4.25–9.79	4	**PASS**
C7d	FpWide<3> inverse >= 5x BigUint modpow(p-2), P-192	16781.68	2402.25	7.634	6.75–9.73	5	**PASS**
C7e	FpWide<4+spare> inverse >= 5x BigUint modpow(p-2), P-224	23205.8	2603.4	8.883	6.66–9.14	5	**PASS**
C8a	Fp32 lazy row update >= 5x eager, p31, 1e4	2.99	0.31	9.453	6.37–11.39	5	**PASS**
C8b	Fp64 lazy row update >= 2x eager, p61, 1e4	7.37	1.87	3.969	3.64–5.08	2	**PASS**
C8c	FpWide<3> lazy row update >= 1.5x eager, P-192, 1e4	20.21	10.46	1.878	1.76–2.3	1.5	**PASS**
C8d	FpWide<3> lazy row update >= 1.5x eager, P-192, 1e5	20.02	10.88	1.823	1.75–2.18	1.5	**PASS**
C8e	FpWide<4+spare> lazy row update >= 1.5x eager, P-224, 1e4	24.72	16.22	1.557	1.37–1.79	1.5	**PASS**
C8f	FpWide<4+spare> lazy row update >= 1.5x eager, P-224, 1e5	29.76	16.54	1.734	1.52–2.26	1.5	**PASS**
C8g	FpWide<4> lazy row update >= 1.5x eager, P-256, 1e4	31.76	16.04	1.992	1.96–2.29	1.5	**PASS**
C9a	FpWide<3> lazy dot >= 1.5x eager, P-192	19.19	7.68	2.575	2.13–3.1	1.5	**PASS**
C9b	FpWide<4+spare> lazy dot >= 1.5x eager, P-224	25.54	12.41	2.093	1.4–2.39	1.5	**PASS**
C10a	binary-GCD (Pornin) inverse >= 1.3x plain binary, FpWide<4>, P-224	4221.22	2630.07	1.576	1.4–1.84	1.3	**PASS**
C10b	binary-GCD (Pornin) inverse >= 2.5x Fermat, FpWide<4>, P-224	8484.14	2630.07	3.29	2.9–4.15	2.5	**PASS**
C11a	binary Jacobi >= 3x Euler, FpWide<3>, P-192	5206.57	1205.72	4.401	3.1–4.92	3	**PASS**
C11b	binary Jacobi >= 3x Euler, FpWide<4>, P-224	8073.5	2187.38	3.609	3.18–5.1	3	**PASS**
C12	windowed Tonelli-Shanks >= 3x plain, FpWide<4>, P-224 (s = 96)	77575.21	21013.3	3.684	2.68–5.44	3	**PASS**
C13	Solinas FpP192 mul >= 1.3x Montgomery FpWide<3>, P-192	18.63	9.94	1.919	1.28–2.59	1.3	**PASS**
C14	spare-bit FpWide<4+spare> mul >= 1.15x FpWide<4>, P-224	28.65	21.06	1.343	1.28–1.67	1.15	**PASS**

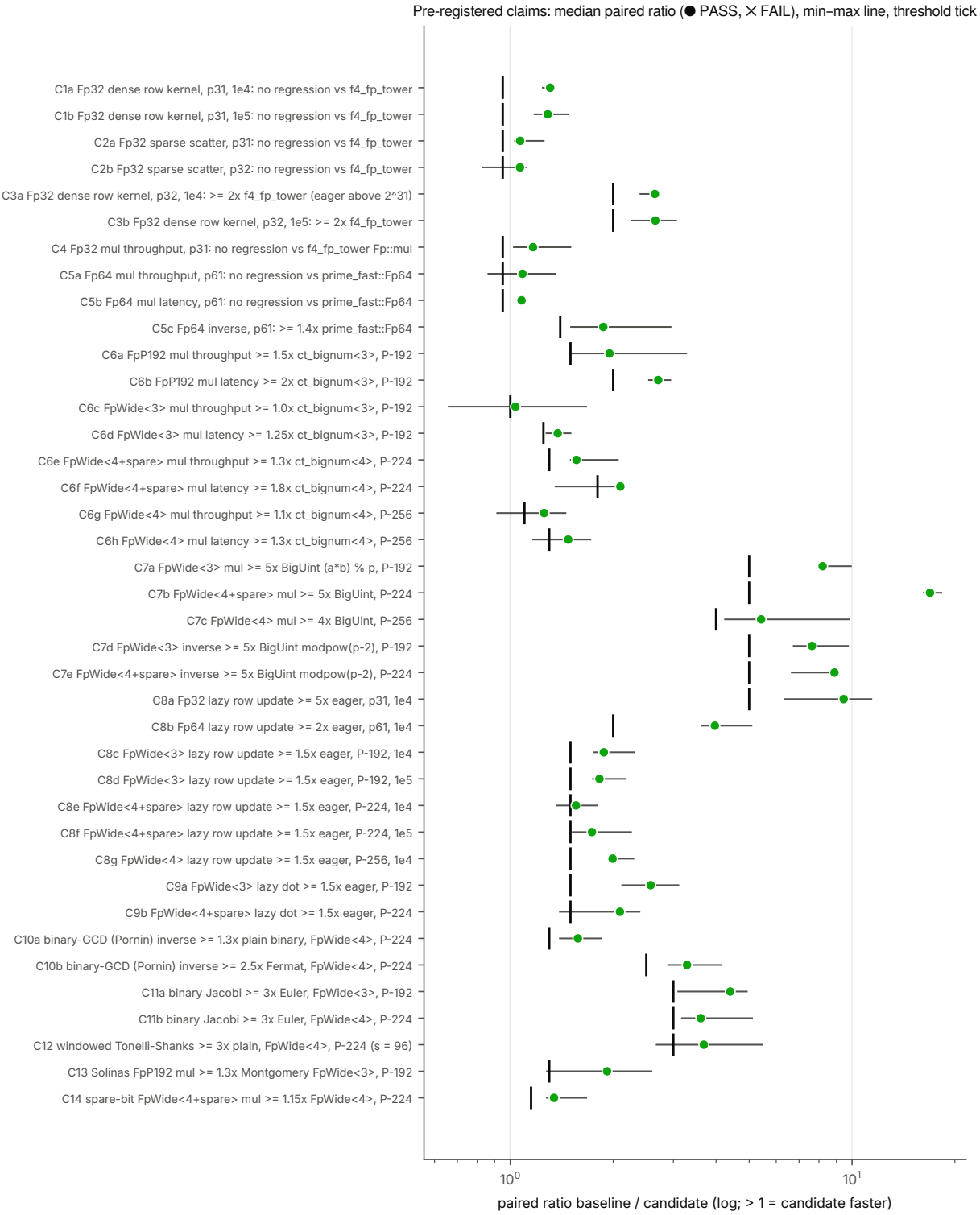


Figure 2: Median paired ratio per claim with min–max over repetitions and the frozen threshold (tick). Run `runs/final`.

All arms (median ns per operation, min, max)

arm	median	min	max
f4_fp_tower/p31/mul_tp	1.88	1.71	2.59
f4_fp_tower/p31/axpy_dense_1e4	0.4	0.39	0.48
Fp32/p31/axpy_dense_1e4	0.31	0.3	0.39
f4_fp_tower/p31/axpy_dense_1e5	0.44	0.41	0.49
Fp32/p31/axpy_dense_1e5	0.34	0.33	0.36
f4_fp_tower/p31/scatter_sparse	2.03	1.89	2.61
Fp32/p31/scatter_sparse	1.92	1.8	2.08
Fp32/p31/scatter_sparse_scalar	2.01	1.9	2.32
f4_fp_tower/p32/mul_tp	1.83	1.64	2.51
f4_fp_tower/p32/axpy_dense_1e4	1.26	1.23	1.46
Fp32/p32/axpy_dense_1e4	0.5	0.46	0.54
f4_fp_tower/p32/axpy_dense_1e5	1.33	1.28	1.54
Fp32/p32/axpy_dense_1e5	0.51	0.48	0.65
f4_fp_tower/p32/scatter_sparse	2	1.94	2.15
Fp32/p32/scatter_sparse	1.86	1.8	2.41
Fp32/p32/scatter_sparse_scalar	3.05	2.98	4.21
Fp32/p31/mul_tp	1.62	1.54	1.77
Fp32/p31/mul_lat	5.17	5.12	5.94
Fp32/p31/sqr_lat	5.21	5.14	6.19
Fp32/p31/add_tp	1.51	1.46	2.45
Fp32/p31/inv	59.34	56.62	78.92
Fp32/p31/batch_inv	10.69	10.57	11.29
Fp32/p31/sqrt	352.03	342.23	451.73
Fp32/p31/legendre	44.32	42.33	53.71
Fp32/p31/axpy_lazy_1e4	0.31	0.3	0.47
Fp32/p31/axpy_eager_1e4	2.99	2.94	3.45
Fp32/p31/axpy_lazy_1e5	0.34	0.33	0.44
Fp32/p31/axpy_eager_1e5	3.04	2.94	3.43
Fp32/p31/dot_lazy_1e4	1.53	1.49	1.64
Fp32/p31/dot_eager_1e4	1.74	1.69	1.84
Fp32/p32/mul_tp	1.7	1.57	1.85
Fp32/p32/mul_lat	5.19	5.13	6.32
Fp32/p32/sqr_lat	5.29	5.12	5.66
Fp32/p32/add_tp	1.39	1.31	2.39
Fp32/p32/inv	63.97	59.74	94.99
Fp32/p32/batch_inv	10.75	10.55	11.97
Fp32/p32/sqrt	190.99	186.45	211.02
Fp32/p32/legendre	43.65	42.34	50.15
Fp32/p32/axpy_lazy_1e4	0.5	0.47	0.54
Fp32/p32/axpy_eager_1e4	2.94	2.92	3.9
Fp32/p32/axpy_lazy_1e5	0.51	0.48	0.65
Fp32/p32/axpy_eager_1e5	2.97	2.95	3.2
Fp32/p32/dot_lazy_1e4	1.14	1.13	1.14
Fp32/p32/dot_eager_1e4	1.71	1.69	1.81
Fp64/p31/mul_tp	2.16	2.11	2.97
Fp64/p31/mul_lat	5.07	4.94	5.38
Fp64/p31/sqr_lat	5.01	4.92	5.71
Fp64/p31/add_tp	1.55	1.34	2.64
Fp64/p31/inv	56.54	52.38	75.61
Fp64/p31/batch_inv	11.33	10.78	13.84
Fp64/p31/sqrt	383.99	364.75	454.88
Fp64/p31/legendre	50	48.1	58.26
Fp64/p31/axpy_lazy_1e4	1.44	1.36	2.05
Fp64/p31/axpy_eager_1e4	7.39	6.95	8.8
Fp64/p31/axpy_lazy_1e5	1.58	1.53	3.11
Fp64/p31/axpy_eager_1e5	7.36	7.23	9.99
Fp64/p31/dot_lazy_1e4	2.14	2.06	2.75
Fp64/p31/dot_eager_1e4	3.44	3.05	5.03
Fp64/p61/mul_tp	2.12	1.98	2.65
Fp64/p61/mul_lat	4.99	4.91	5.19
Fp64/p61/sqr_lat	4.99	4.94	5.12
Fp64/p61/add_tp	1.33	1.29	2.09
Fp64/p61/inv	245.67	234.49	288.68
Fp64/p61/batch_inv	10.93	10.81	11.37
Fp64/p61/sqrt	293.63	287.6	304.47
Fp64/p61/legendre	90.94	89.49	98.39
Fp64/p61/axpy_lazy_1e4	1.87	1.77	2
Fp64/p61/axpy_eager_1e4	7.37	6.99	9.08
Fp64/p61/axpy_lazy_1e5	2.04	1.91	3.3
Fp64/p61/axpy_eager_1e5	7.81	7.49	9.59
Fp64/p61/dot_lazy_1e4	2.12	2.08	2.53
Fp64/p61/dot_eager_1e4	3.21	3.02	3.97
Fp64/p63/mul_tp	2.07	1.99	2.28
Fp64/p63/mul_lat	4.99	4.93	5.61
Fp64/p63/sqr_lat	5.01	4.93	5.13

arm	median	min	max
Fp64/p63/add_tp	1.34	1.27	2.4
Fp64/p63/inv	267.51	249.39	317.39
Fp64/p63/batch_inv	10.91	10.7	11.3
Fp64/p63/sqrt	399.76	389.41	522.59
Fp64/p63/legendre	96.11	92.61	102.59
Fp64/p63/axpy_lazy_1e4	2.65	2.52	3.47
Fp64/p63/axpy_eager_1e4	7.51	7.27	8.23
Fp64/p63/axpy_lazy_1e5	3.21	3	3.62
Fp64/p63/axpy_eager_1e5	8	7.77	9.6
Fp64/p63/dot_lazy_1e4	2.09	2.06	2.21
Fp64/p63/dot_eager_1e4	3.92	3.77	4.72
prime_fast::Fp64/p61/mul_tp	2.22	2.15	2.68
prime_fast::Fp64/p61/mul_lat	5.35	5.31	5.6
prime_fast::Fp64/p61/inv	455.34	434.62	710.42
FpWide<3>/P-192/mul_tp	18.63	17.43	28.33
FpWide<3>/P-192/mul_lat	21.93	21.54	23.81
FpWide<3>/P-192/sqr_lat	20.51	20.15	22.33
FpWide<3>/P-192/add_tp	5	4.06	6.69
FpWide<3>/P-192/inv	2402.25	2151.05	3080.4
FpWide<3>/P-192/batch_inv	54.68	52.64	79.76
FpWide<3>/P-192/sqrt	4884.6	4804.84	5928.3
FpWide<3>/P-192/legendre	1205.72	1141.35	1639.6
FpWide<3>/P-192/axpy_lazy_1e4	10.46	10.21	11.47
FpWide<3>/P-192/axpy_eager_1e4	20.21	18.64	23.49
FpWide<3>/P-192/axpy_lazy_1e5	10.88	10.6	11.8
FpWide<3>/P-192/axpy_eager_1e5	20.02	18.84	23.33
FpWide<3>/P-192/dot_lazy_1e4	7.68	7.2	9.14
FpWide<3>/P-192/dot_eager_1e4	19.19	18.39	28.39
FpP192/P-192/mul_tp	9.94	9.15	13.91
FpP192/P-192/mul_lat	11.15	11.06	12.05
FpP192/P-192/sqr_lat	10.19	10.06	10.85
FpP192/P-192/add_tp	4.18	4.09	5.82
FpP192/P-192/inv	2114.67	2054.65	2746.56
FpP192/P-192/batch_inv	32.14	31.67	42.42
FpP192/P-192/sqrt	2502.6	2392.36	2868.31
FpP192/P-192/legendre	1235.99	1074.14	1702.97
FpP192/P-192/axpy_lazy_1e4	9.75	9.12	11.09
FpP192/P-192/axpy_eager_1e4	12.07	11.68	16.09
FpP192/P-192/axpy_lazy_1e5	10.08	9.94	12.87
FpP192/P-192/axpy_eager_1e5	12.24	11.88	13.03
FpP192/P-192/dot_lazy_1e4	7.37	7	9.75
FpP192/P-192/dot_eager_1e4	12.12	11.66	13.62
FpWide<4>/p200/mul_tp	31.44	29.05	38.56
FpWide<4>/p200/mul_lat	30.79	29.45	39.56
FpWide<4>/p200/sqr_lat	29.08	28.96	33.62
FpWide<4>/p200/add_tp	5.5	5.23	6.89
FpWide<4>/p200/inv	2349.76	2274.57	2774.16
FpWide<4>/p200/batch_inv	86.5	79.48	94.05
FpWide<4>/p200/sqrt	6422.44	6302.41	7693.59
FpWide<4>/p200/legendre	1885.39	1806	2421.76
FpWide<4>/p200/axpy_lazy_1e4	17.09	15.95	20.9
FpWide<4>/p200/axpy_eager_1e4	33.75	30.87	48.64
FpWide<4>/p200/axpy_lazy_1e5	17.93	16.35	22.31
FpWide<4>/p200/axpy_eager_1e5	33.47	31.71	36.63
FpWide<4>/p200/dot_lazy_1e4	11.96	11.42	22.15
FpWide<4>/p200/dot_eager_1e4	33.9	30.87	51.54
FpWide<4>/P-224/mul_tp	28.65	26.66	36.13
FpWide<4>/P-224/mul_lat	31.73	29.21	39.73
FpWide<4>/P-224/sqr_lat	30.12	28.57	35.14
FpWide<4>/P-224/add_tp	5.62	5.02	7.91
FpWide<4>/P-224/inv	2630.07	2562.73	2857.09
FpWide<4>/P-224/batch_inv	94.57	79.93	128.76
FpWide<4>/P-224/sqrt	21013.3	18754.21	30317.65
FpWide<4>/P-224/legendre	2187.38	2088.66	2793.78
FpWide<4>/P-224/axpy_lazy_1e4	16.36	16.14	17.12
FpWide<4>/P-224/axpy_eager_1e4	32.41	31.08	40.11
FpWide<4>/P-224/axpy_lazy_1e5	17.18	16.81	20.21
FpWide<4>/P-224/axpy_eager_1e5	32.59	32.06	35.39
FpWide<4>/P-224/dot_lazy_1e4	11.78	11.44	17.13
FpWide<4>/P-224/dot_eager_1e4	31.55	31.05	39.37
FpWide<4+spare>/p200/mul_tp	22.4	20.79	39.45
FpWide<4+spare>/p200/mul_lat	21.23	20.98	23.52
FpWide<4+spare>/p200/sqr_lat	20.34	19.8	31.07
FpWide<4+spare>/p200/add_tp	5.26	5.09	9.73
FpWide<4+spare>/p200/inv	2344.75	2274.03	3302.29
FpWide<4+spare>/p200/batch_inv	66.47	61.64	114.38
FpWide<4+spare>/p200/sqrt	4584.84	4467.28	7452.39

arm	median	min	max
FpWide<4+spare>/p200/legendre	1857.05	1803.68	2561.84
FpWide<4+spare>/p200/axpy_lazy_1e4	15.62	15.04	28.13
FpWide<4+spare>/p200/axpy_eager_1e4	24.11	23.78	43.85
FpWide<4+spare>/p200/axpy_lazy_1e5	16.72	15.95	28.11
FpWide<4+spare>/p200/axpy_eager_1e5	25.71	24.29	42.95
FpWide<4+spare>/p200/dot_lazy_1e4	11.92	11.35	12.82
FpWide<4+spare>/p200/dot_eager_1e4	24.04	23.11	28.88
FpWide<4+spare>/P-224/mul_tp	21.06	20.81	24.02
FpWide<4+spare>/P-224/mul_lat	21.39	21.01	32.56
FpWide<4+spare>/P-224/sqr_lat	22.58	19.79	28.88
FpWide<4+spare>/P-224/add_tp	5.3	5.11	8.85
FpWide<4+spare>/P-224/inv	2603.4	2552.91	3678.53
FpWide<4+spare>/P-224/batch_inv	65.87	63.54	80.22
FpWide<4+spare>/P-224/sqrt	14432.28	13735.65	21724.74
FpWide<4+spare>/P-224/legendre	2163.56	2086.93	2667.07
FpWide<4+spare>/P-224/axpy_lazy_1e4	16.22	15.33	18.25
FpWide<4+spare>/P-224/axpy_eager_1e4	24.72	23.91	32.74
FpWide<4+spare>/P-224/axpy_lazy_1e5	16.54	15.81	23.01
FpWide<4+spare>/P-224/axpy_eager_1e5	29.76	24.28	51.91
FpWide<4+spare>/P-224/dot_lazy_1e4	12.41	11.49	19.1
FpWide<4+spare>/P-224/dot_eager_1e4	25.54	23.59	40.4
FpWide<4>/P-256/mul_tp	28.68	26.9	36.16
FpWide<4>/P-256/mul_lat	29.96	29.26	38.36
FpWide<4>/P-256/sqr_lat	29.07	28.61	30.18
FpWide<4>/P-256/add_tp	5.07	4.95	5.84
FpWide<4>/P-256/inv	2999.54	2918.2	3087.35
FpWide<4>/P-256/batch_inv	79.77	78.84	112.49
FpWide<4>/P-256/sqrt	7914.1	7804.25	9214.57
FpWide<4>/P-256/legendre	2538.62	2426.66	3518.99
FpWide<4>/P-256/axpy_lazy_1e4	16.04	15.45	21.99
FpWide<4>/P-256/axpy_eager_1e4	31.76	30.61	46.37
FpWide<4>/P-256/axpy_lazy_1e5	17.02	16.11	20.8
FpWide<4>/P-256/axpy_eager_1e5	35	31.18	50.13
FpWide<4>/P-256/dot_lazy_1e4	13.22	11.39	17.94
FpWide<4>/P-256/dot_eager_1e4	33.32	30.9	49.55
ct_bignum/P-192/mul_tp	18.86	18.36	32.53
ct_bignum/P-192/mul_lat	30.29	29.68	32.91
ct_bignum/P-192/sqr_lat	30.75	30.37	32.13
BigUint/P-192/mul_tp	152.81	146.13	227.23
BigUint/P-192/inv	16781.68	16265.73	25098.5
ct_bignum/P-224/mul_tp	34.26	31.94	43.38
ct_bignum/P-224/mul_lat	45.17	43.99	47.44
ct_bignum/P-224/sqr_lat	44.19	43.47	45.4
BigUint/P-224/mul_tp	356.77	341	395.01
BigUint/P-224/inv	23205.8	22607.64	24498.46
ct_bignum/P-256/mul_tp	33.98	32.45	46.04
ct_bignum/P-256/mul_lat	44.27	43.82	51.87
ct_bignum/P-256/sqr_lat	44.16	43.75	45.79
BigUint/P-256/mul_tp	164.41	153.64	263.33
BigUint/P-256/inv	26585.56	25888.95	38417.73
FpWide<3>/P-192/inv_fermat	5236.58	5172.88	5498.13
FpWide<3>/P-192/inv_plain_binary	2785.34	2730.96	2997.74
FpWide<3>/P-192/legendre_euler	5206.57	5082.48	5962.41
FpWide<4>/P-224/inv_fermat	8484.14	8273.58	11182.46
FpWide<4>/P-224/inv_plain_binary	4221.22	3990.21	4719.99
FpWide<4>/P-224/legendre_euler	8073.5	7497.28	11154.57
FpWide<4>/P-224/sqrt_ts_plain	77575.21	75615.26	110226.91